Cài đặt Apache Spark

1. Tìm hiểu lý thuyết Apache Spark

https://www.tutorialspoint.com/apache_spark/index.htm

2. Cài đặt Scala

apt-get install scala

scala



Thoát scala

```
sys.exit
```

3. Cài đặt Spark

$ wget https://dlcdn.apache.org/spark/spark-3.2.1/spark-3.2.1-bin-hadoop3.2.tgz

Khai báo ~/.bashrc

Khởi động Spark

$spark-shell

```
hadoopminhchau@minhchau-master:~$ spark-shell
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.spark.unsafe.Platform (file:/home/hadoopminhchau/
ark/jars/spark-unsafe_2.12-3.2.1.jar) to constructor java.nio.DirectByteBuffer(long,int)
WARNING: Please consider reporting this to the maintainers of org.apache.spark.unsafe.Platform
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operat
s
WARNING: All illegal access operations will be denied in a future release
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
2022-03-30 10:46:54,917 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your
atform... using builtin-java classes where applicable
Spark context Web UI available at http://minhchau-master:4040
Spark context available as 'sc' (master = local[*], app id = local-1648637215825).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.2.1
      /_/

Using Scala version 2.12.15 (OpenJDK 64-Bit Server VM, Java 11.0.14)
Type in expressions to have them evaluated.
Type :help for more information.

scala> _
```

examples/src/main/scala/org/apache/spark/examples/SparkPi.scala

$run-example SparkPi 10